

**ASSESSMENT OF GROUNDWATER QUALITY OF KANAGANAPALLI MANDAL,
ANANTHAPURAMU DISTRICT, BY USING GEOSPATIAL TECHNIQUES**

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ABSTRACT

Groundwater is one of the important freshwater resources accessible for human utilization. However, evaluation of the quality of groundwater is necessary to improve its portability. The present study is aimed to assess the groundwater quality of Kanaganapalli Mandal, Anantapur District of Andhra Pradesh State, India. Twenty-two groundwater samples were collected from dissimilar locations of the study area per the period of post-monsoon December (2020) and locations are recognized with GPS. The groundwater samples analyzed in physical and chemical properties such as pH, EC, TDS, major Cations, and major anions were measured using standard procedure. The results obtained were found to exceed the permissible limits. It is observed that most of the groundwater is alkaline in nature. The Total Dissolved Solids of the groundwater in the study area is ranging from 350 mg/l to 1840 mg/l. The calcium concentration of the groundwater in the study area is ranging from 60 mg/l to 140 mg/l. The chloride concentration of the groundwater in the study area is ranging from 213 to 401 mg/l during the post-monsoon period. The Total hardness of the study area is ranging from 90 mg/l to 310 mg/l. The bicarbonate concentration of the groundwater in the study area is ranging from 300 mg/l to 600 mg/l during the post-monsoon period. The Groundwater of the study area is very hard and the Mg concentration of the groundwater in the study area is ranging from 3 mg/l to 41 mg/l. All parameters except pH and EC indicate the unit of Mg/l. all the measured parameters are compared with the WHO (2014) standard in favor of drinking water quality permissible limit. In this case, the study area samples are occurred some samples above and below the limit, which is the effect of some anthropogenic activities.

KEYWORDS: Groundwater, Hydrogeochemistry, Kanaganapalli, and WHO (World health organization, Spatial distribution Maps.

INTRODUCTION

Water is an elixir of life. Life, prosperity revolves around water on the whole planet. Groundwater plays a vital role in the world economy. About 64% of rural and 37% urban Indians are without the right to use safe drinking water as per the WHO (Aghazadeh et al. 2010; Balakrishnan et al. 2011; Golla et al. 2021). Life is not possible without water; because water is the most indispensable resource. Water is an inorganic, neutral chemical material, which is the chief element of the earth's hydrosphere. It is a chemical complex and may occur in a fluid or solid or gaseous state. All these three states of water are tremendously helpful for people, give luxuries and relieves. It is broadly utilized in industrialized methods and agriculture. This study area used GIS as a database structure in arrange to prepare maps of water superiority (Islam et al.2013; Krupavathi et al. 2020; Kumar et al. 2020; Badapalli et al. 2020) Groundwater quality map is vital to estimate the water safeness for drinking and agriculture reasons and also as a deterrent warning to the environmental potential and physical condition. The study area is using geospatial techniques in groundwater quality mapping and present study was taken up to map the groundwater superiority in Anantapur district, Andhra Pradesh, India. In this study area, the main objective of research work is to make a groundwater quality assessment using Geospatial techniques (Islam et al. 2014; Nagaraju et al. 2014; Ibraheem et al; 2017; Kumar et al. 2020).

STUDY AREA

The study area is situated south of Anantapur district, Andhra Pradesh, India, it is enclosed 503 sq km. This study area geographically lies between 14° 26'58" and 14° 27' 0" N latitude and 77°32'2" and 77°31'1" E longitudes (Fig.1). It has an average elevation of 426 meters (1400 ft). The study area falls in the Survey of India Toposheet Nos. 57F/6, F/7, and F/11. The study area geologically comprises of Hornblende Biotite Gneiss, Meta basalts, Grey/pink granites, Granites, Granodiorite, and Quartzites (Fig. 2). Twenty-two water sample location sites along with their latitude and longitudes area shown in Table (1).

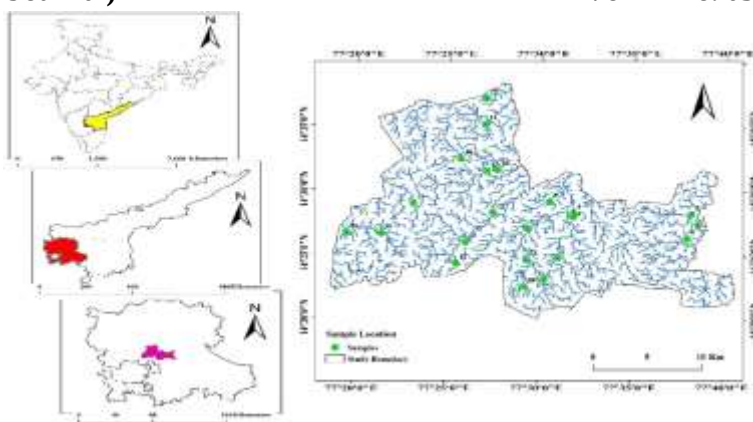


Figure 1: Sample location map of the Study area

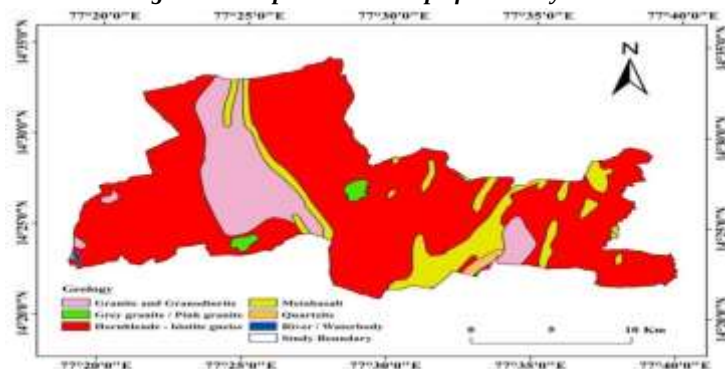


Figure 2: Geology map of the study area

Table: 1 Water sample locations

S. No	Sample ID	Name of the village	Latitude	Longitude
1	K1	Mavillapalle	N 14°26'21"	E 77°38'3"
2	K2	Mukthapuram	N 14°26'58"	E 77°37'42"
3	K3	Dadalur	N 14°26'46"	E 77°31'33"
4	K4	Kanaganapalli (a)	N 14°26'58"	E 77°31'3"
5	K5	Kanaganapalli (b)	N 14°29'24"	E 77°33'52"
6	K6	Ellakuntla	N 14°26'1"	E 77°29'8"
7	K7	Balepalem	N 14°24'26"	E 77°30'48"
8	K8	Nemalivaram	N 14°25'35"	E 77°37'28"
9	K9	Thallimadugula	N 14°23'12"	E 77°30'4"
10	K10	Muthuvakuntla	N 14°22'45"	E 77°29'3"
11	K11	Gudivandlapalli	N 14°24'3"	E 77°25'30"
12	K12	Narasampalli	N 14°25'16"	E 77°25'58"
13	K13	Baddalapuram	N 14°26'57"	E 77°26'57"
14	K14	Thumcherla	N 14°31'57"	E 77°26'56"
15	K15	C.N.Kunta	N 14°33'24"	E 77°27'55"
16	K16	Thogarakunta	N 14°29'24"	E 77°27'56"
17	K17	K.K.Thanda	N 14°29'18"	E 77°27'32"
18	K18	Guntapalli	N 14°29'6"	E 77°25'42"
19	K19	Vepakunta	N 14°27'3"	E 77°33'30"
20	K20	Maddalachervu	N 14°27'25"	E 77°26'35"
21	K21	M.C.Thanda	N 14°25'43"	E 77°26'30"
22	K22	Somavarandlapalli	N 14°24'18"	E 77°29'81"

METHODOLOGY

Twenty-two groundwater samples were collected from open and dug wells from various villages of Kanaganapalli Mandal of Anantapur District for the period of post-monsoon in the month of December 2020. The groundwater samples were collected in well-cleaned 1000 ml polyethylene bottles and they have been analyzed in the laboratory and the results of the analysis of water samples parameters are shown in Table 2 & 3. The latitude and longitude of the sample locations are identified with a Global positioning system (GPS) (Srinivasa Gowd, 2005; Singaraja et al. 2015). The methodology of the study region is shown in Fig. 3.

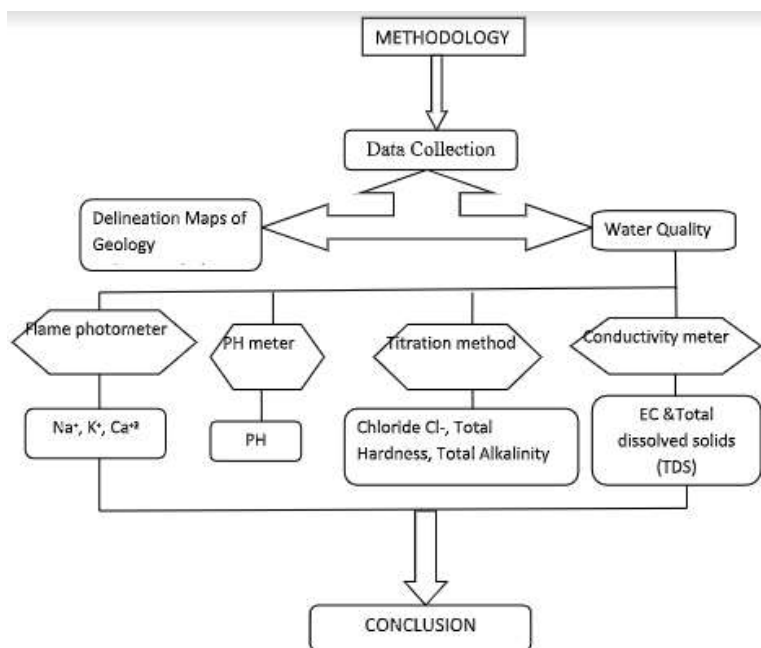


Figure 3: Methodology of the study area

Table 2: Results of the physico chemical parameters of the Kanaganapalli Mandal.

S.No	Sample ID	TH	SO ₄	Cl	CO ₃	HCO ₃	pH	EC	TDS	Ca ⁺²	Mg ⁺²	Na ⁺	K ⁺
		Mg/l	Mg/l	Mg/l	Mg/l	Mg/l		us/m	Mg/l	Mg/l	Mg/l	Mg/l	Mg/l
1	K1	100	21	258	18	159	7.8	1134	580	60	12	142	6
2	K2	250	40	273	30	183	7.8	2643	1360	140	24	126	5
3	K3	100	50	365	30	85	7.7	1435	780	88	29	134	7
4	K4	120	30	254	18	122	7.7	1127	654	84	22	165	4
5	K5	100	40	368	24	122	7.8	1232	715	88	29	158	5
6	K6	90	65	298	12	110	7.7	2122	1125	100	39	119	4
7	K7	240	54	238	36	98	7.8	1192	727	130	18	186	5
8	K8	220	48	401	-	122	7.8	3193	1640	76	7	128	6
9	K9	250	65	213	60	146	7.7	3600	1840	104	4	175	6
10	K10	110	55	267	30	110	7.8	2450	1186	100	3	198	7
11	K11	120	36	231	30	183	7.5	2912	1516	108	36	212	4
12	K12	150	65	297	30	61	7.6	3000	1534	80	12	183	5

13	K13	150	54	276	18	73	7.5	951	485	80	12	178	7
14	K14	100	35	298	30	146	7.6	946	481	88	29	156	5
15	K15	130	60	280	54	98	7.6	1560	770	68	10	168	6
16	K16	100	80	365	24	61	7.7	1173	596	88	12	98	6
17	K17	90	40	386	18	61	7.9	1660	858	76	24	196	7
18	K18	310	75	398	18	122	7.7	687	350	64	36	259	6
19	K19	180	40	296	12	195	7.8	1064	437	104	29	276	6
20	K20	110	80	365	18	122	7.6	1114	558	72	17	196	7
21	K21	120	40	376	20	146	7.8	875	797	76	17	274	5
22	K22	130	99	398	36	195	7.6	2985	1428	104	41	179	7.8

Table 3: Cation & Anion Balance of groundwater of Study area

S.No	Sample ID	SO ₄	Cl	CO ₃	HCO ₃	Total Anions	Ca	Mg	Na	K	Total cations
		Meq/l	Meq/l	Meq/l	Meq/l		Meq/l	Meq/l	Meq/l	Meq/l	
1	K1	0.4164	7.278	0.5999	2.606	10.9003	2.994	1.729	5.22	0.1534	10.0964
2	K2	0.8328	7.701	0.9999	2.9993	12.533	6.986	3.458	2.17	0.1278	12.7418
3	K3	1.041	10.296	0.9999	1.3931	13.73	4.391	4.179	4.785	0.1789	13.5339
4	K4	0.6246	7.165	0.5999	1.9995	10.389	4.191	3.1704	3.262	0.1022	10.7256
5	K5	0.8328	10.381	0.7999	1.9995	14.0132	4.391	4.1791	5.481	0.1278	14.1789
6	K6	1.3533	9.732	0.3999	1.8029	13.2885	4.99	5.62	2.392	0.1022	13.1042
7	K7	1.1242	8.406	1.1998	1.6062	12.3362	6.487	2.593	3.48	0.12785	12.68785
8	K8	0.9993	10.71	-	1.9995	13.7088	3.792	1.008	8.482	0.1534	13.4354
9	K9	1.3533	6.008	1.1998	2.3929	10.954	5.189	0.4323	3.915	0.1534	9.6897
10	K10	1.1451	5.952	0.9999	1.8029	9.8999	4.99	0.4323	4.132	0.1789	9.7332
11	K11	0.7495	9.027	0.9999	2.9993	13.7757	5.389	5.1879	3.262	0.1022	13.9411
12	K12	1.3533	8.378	0.9999	1.1964	11.9276	3.992	1.7293	5.437	0.1278	11.2861
13	K13	1.1242	7.983	0.5994	2.3929	12.0995	3.992	4.1791	3.654	0.1789	12.004
14	K14	0.7282	8.405	0.9999	1.6062	11.7393	4.391	1.4411	5.22	0.1278	11.1799
15	K15	1.2492	6.629	1.7998	0.9997	10.6777	3.393	1.4411	5.665	0.1534	11.4947
16	K16	1.6656	8.152	0.7999	0.9997	11.6172	4.392	1.7293	5.22	0.1534	11.4947
17	K17	0.8328	8.18	0.5999	1.9995	11.6122	3.7924	3.458	3.871	0.1789	11.3003
18	K18	0.5615	8.03	0.5999	3.196	12.3874	3.1936	5.1879	2.61	0.1534	11.1449
19	K19	0.8328	8.35	0.3999	1.9995	11.5822	3.59	4.1791	3.741	0.1534	11.6635
20	K20	0.4164	8.35	0.5999	1.9995	11.3658	3.592	2.4498	4.78	0.1789	11.0007
21	K21	0.8328	6.601	0.6666	2.3929	10.4933	3.7924	2.4498	3.74	0.1278	10.11
22	K22	2.082	9.196	1.1998	3.19605	15.67385	5.1896	5.9085	4.132	0.2045	15.4346

RESULTS AND DISCUSSION

Hydrogen Ion Concentration (pH)

The pH determined the concentrations of the hydrogen ion $[H^+]$ in the study area. pH is the largest vital indicator of groundwater quality and provides a significant portion of information in numerous categories of geochemical stability or solubility estimates. In general, up to a standard range of pH is 6.5 to 8.5 (WHO, 2011). In nearly everyone's natural water; the pH dependent on the carbonate, carbon dioxide, and bicarbonate balance. As the balance is noticeably exaggerated by temperature and pressure, it is understandable that modify in pH could take place at what time these are distorted. The majority of the groundwater has a pH range of 6.5-8.5. The pH of the study area ranges from 7.5 to 7.9 (Fig. 4). pH values are vital in medicine, living sciences, and many other applications [9]. The pH values for all the samples are contained by the desirable limit. All samples specify a slightly alkaline nature. A spatial distribution map is also prepared for the pH values which are shown in Fig 5.

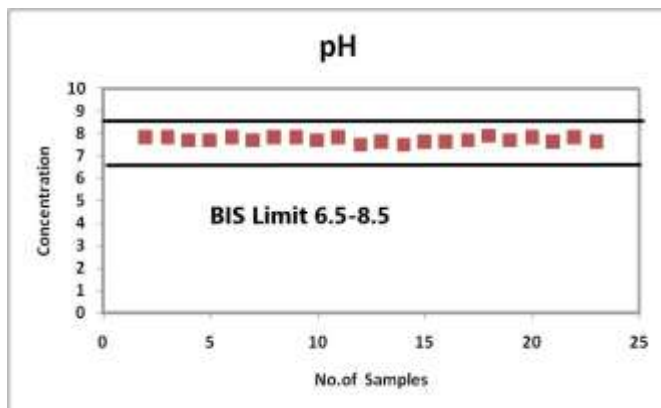


Figure 4: Graphical representation of pH of the study area

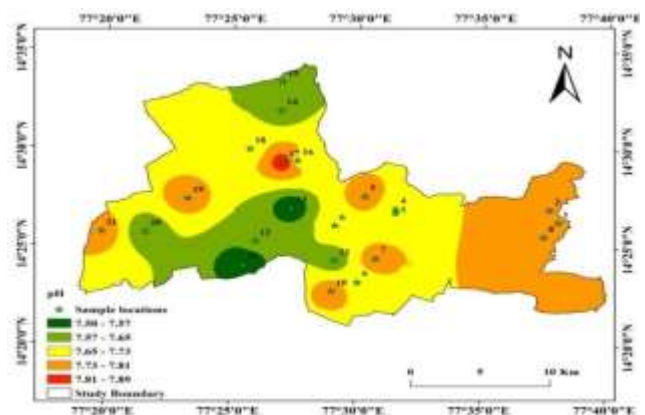


Figure 5: Distribution of pH of the Study area

Total Hardness

Total hardness is described as the number of suspended materials of calcium and magnesium in the water. Hard water consists of suspended materials that are high, such as calcium and magnesium. Total hardness in the study area is ranging from 90 to 310mg/l (Fig. 6). The limit of total hardness for drinking water is particular as 300 mg/l. Water hardness is chiefly due to the end result of contact between water and the geological formations. The Groundwater of the whole study area goes over the acceptable limits. The maximum values of total hardness distinguished at Guntapalli and the minimum values of total hardness distinguished at Ellakuntla. A spatial distribution map is also prepared for the TH values which are shown in Fig 7. From the map, it is observed that the TH values are well within the permissible limit [10].

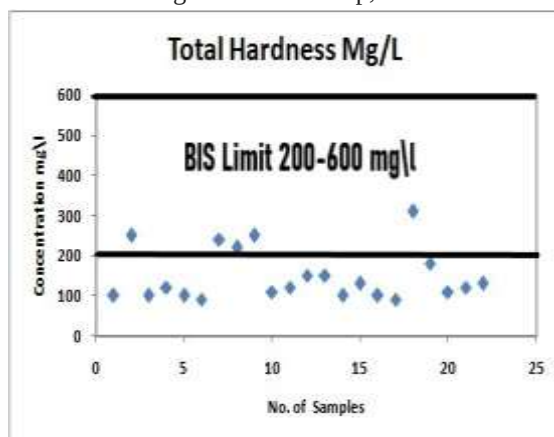


Figure 6: Graphical representation of pH in Groundwater of the study area

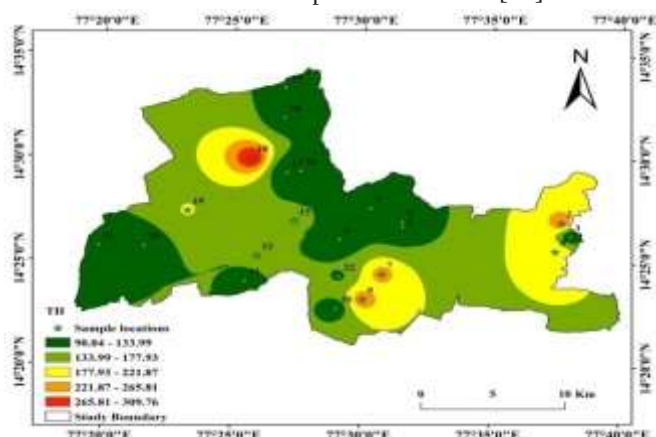


Figure 7: Distribution of Total hardness in Groundwater of the study area

Classification of Ground water based on Total hardness

Parameter	Classification	Range
Total Hardness	Safe	<75
	Moderate hard	75-150
	Hard	150-300
	Very hard	>300

Total Dissolved Solid (TDS)

Total dissolved solids (TDS) is a word utilized to explain the inorganic salts and little quantity of organic matter present in samples (Nagaraju et al., 2014). The attentiveness of suspending materials are vital factors for drinking purpose so as to provides a specific taste to the water a superior concentration and decreases its portability. Maximum concentrations of suspending materials 3000 mg/l could too produce distress in livestock (Punam Tyagi, 2003). From the total dissolved solids analysis, it has been found that the values are ranging between 350 to 1840 mg/l (Fig. 8). The lowest dissolved solids values of 350 mg/l distinguished in Guntapalli, the highest total dissolved solids value of 1840 mg/l is distinguished at Thallimadugula. A spatial distribution map is also prepared for the TDS values which are shown in Fig (9). In this map shown TDS values are within the permissible limit.

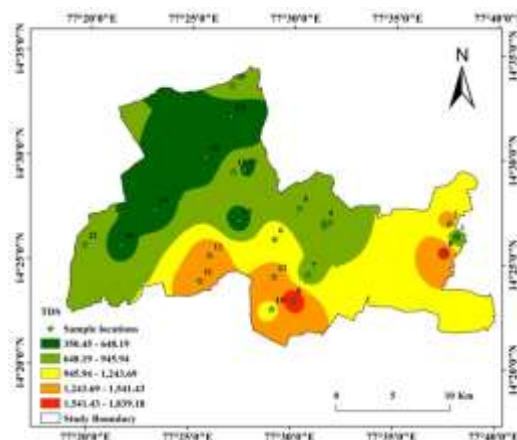


Figure 8: Distribution of TDS in Groundwater of the study area

ELECTRICAL CONDUCTIVITY

Electrical conductivity is an excellent measurement of salinity hazards to crops. It depends upon temperature, concentration, and a lot of ions are situated. Consequently, as the concentration of dissolved salts increases, it gradually increases conductivity too. Alkalies chlorides, sulfides, and carbonate compounds are source for conductive ions. In this study area, experimental results include experimental ranges from 687 μ mho/cm to 3600 μ mho/cm. The least value is observed in Guntapalli and the greatest value is observed in Thallimadugula. EC values which are shown in a spatial distribution map Fig 9.

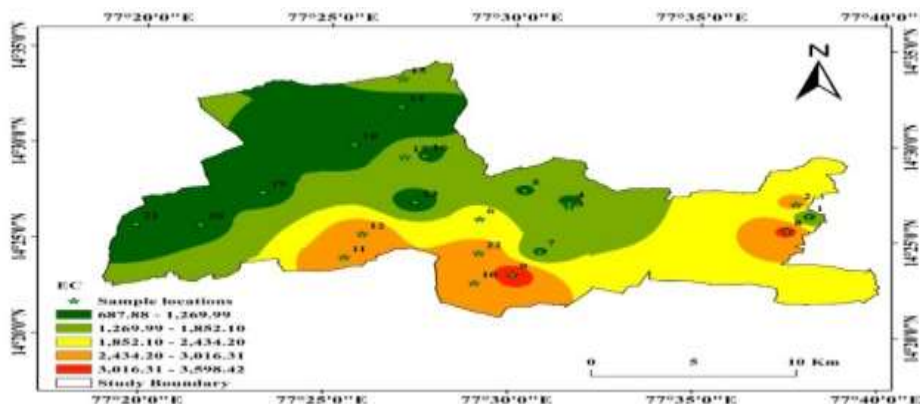


Figure 9: Distribution of EC in Groundwater of the study area

ANIONS

CHLORIDE

Chloride is a naturally occurring frequently in natural waters and is the majority regularly found as a component of salt or in some cases in a mixture of potassium and calcium (Srinivasa Gowd 2005). The study area consists of Chloride concentrations ranges from 213mg/l to 401 mg/l (Fig. 10). The Maximum concentration of chlorides distinguished at Nemalivaram, Minimum concentration of chlorides distinguished at Thallimadugula. The up to the standard limit is chloride for drinking water is particular as 250 mg/l to 600mg/l (WHO 2011). The Cl⁻ concentration is shown in a spatial distribution map Fig 11. This map shows Cl⁻ values are within the acceptable limit.

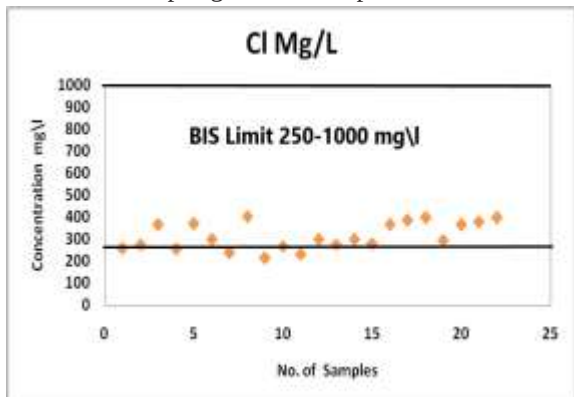


Figure 10: Graphical representation of Cl⁻ in Groundwater of the study area

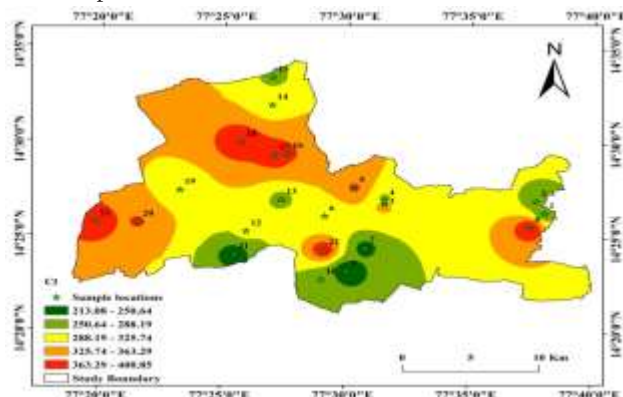


Figure 11: Distribution of Cl⁻ of the study area

TOTAL ALKALINITY (CO₃ AND HCO₃)

Alkalinity is a water acid-neutralizing capability and is the principal function of carbonates and hydroxide content. Bicarbonates are chiefly formed by alkalinity. Carbonates and Bicarbonates are extremely present in the groundwater due to the reaction between water, calcium, and magnesium in the form of a solution into the soil. The allowable limit bicarbonate (HCO₃) 500 mg/l in the study area (Standard, I. 2012; Mynar Babu et al. 2012; Ravi Kumar et al. 2020). The bicarbonate concentration ranges from 61 mg/l to 195 mg/l. The suitable limit of carbonate (CO₃) in drinking water is 10 mg/l and the unsuitable limit is 50 mg/l. The carbonate concentration ranges from 0 to 60 mg/l. in the study region. The minimum concentration of Bicarbonates observed in Thogarakunta. The maximum Concentration of Bicarbonates was observed in Vepakunta & Somavarandlapalli. The HCO₃⁻ concentration which is shown in a spatial distribution map Fig 12. In this map shown the HCO₃⁻ ranges are within the permissible limit.

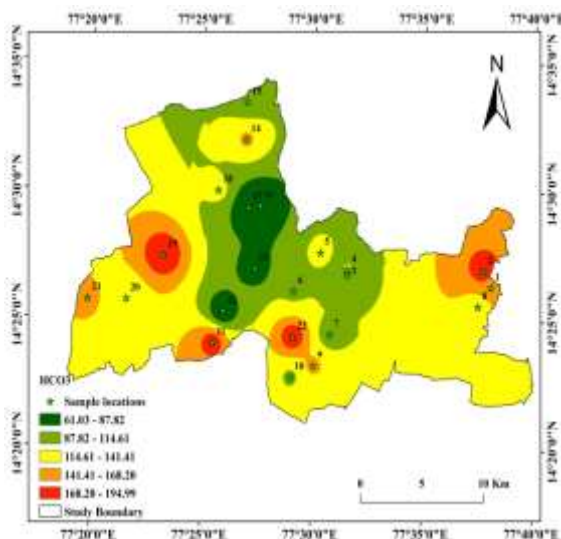


Figure 12: Distribution of HCO₃⁻ in Groundwater of the study area.

SULFATE

Sulfate is one of the chiefly suspend elements of rain. A high concentration of SO_4^- in the groundwater reasons to laxative effect at what time together with magnesium and calcium, these are the elements of hardness. Concentrations in excess of 500mg/l especially if the magnesium content is also high may have a laxative effect or gastrointestinal irritation. In the study region sulfate, Concentration ranges from 21mg/l to 99mg/l. A high concentration of sulfate noted at Somavarandlapalli, low concentration of sulfate noted at Mavillapalle. The acceptable limit of sulfate concentration for drinking water is particular at 200mg/l (WHO 2011). The SO_4^{2-} concentration is shown in a spatial distribution map Fig 13. From the map, it is shown the Sulfate ranges are within the permissible limit.

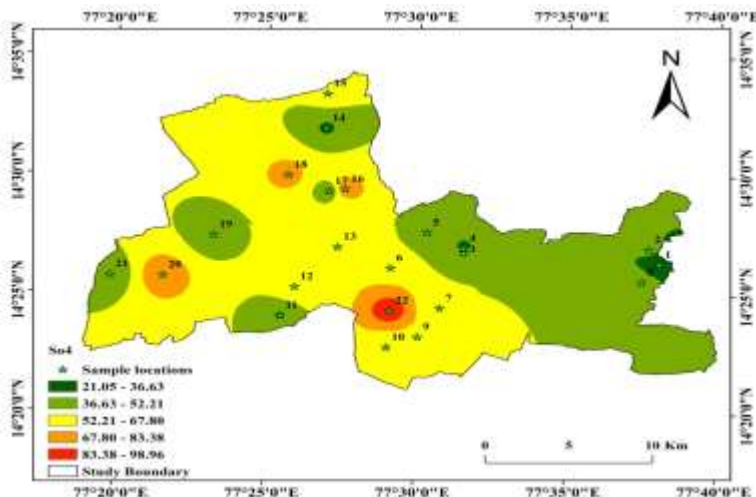


Figure 13: Distribution of sulfate in Groundwater of the study area

CATIONS

Calcium

Calcium has naturally occurred in water. It could suspend from geological formations like calcite limestone, dolomite, marble, gypsum, apatite and fluorite. Calcium is defined as water hardness because it is able to originate in water as Ca^{2+} ions. Even although the human body has a need of roughly 0.7 to 2.0 grams of calcium per day as food stuff constituent, extreme quantity knows how to show the way to the configuration of gallbladder stones. Calcium serves a vital role in human bodies of water. The up to the standard limit is 75mg/l to 200mg/l (WHO 2011). The Calcium concentration of the study area ranges from 60mg/l to 140mg/l (Fig.14). The minimum concentration of calcium reported in Mavillapalle and the maximum concentration of calcium reported in Mukthapuram. The Ca^{2+} concentration is shown in a spatial distribution map Fig 15. These maps show the Ca^{2+} ranges are within the permissible limit.

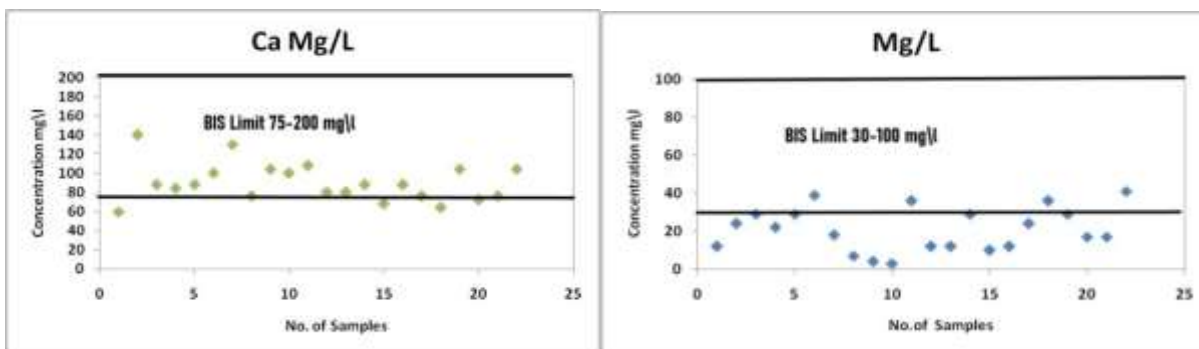


Figure 14: Graphical representation of Ca^{2+} and Mg^{2+} in Groundwater of the study area

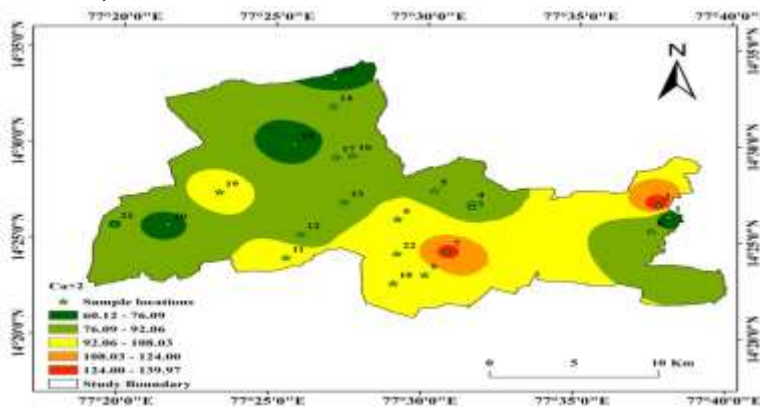


Figure 15: Distribution of Ca^{2+} in Groundwater of the study area

MAGNESIUM

Magnesium is a vital component of basic igneous rocks such as pyroxenites, dunites, and amphibolites; basalts, tremolite-schists and talc; and dolomite; biotite augite, olivine, serpentine, hornblende, and talc are chiefly consisting of magnesium-bearing minerals. Magnesium ions are a specific consequence of water contamination. They can give water hardness. Concentrations of magnesium and calcium in the water could too be a feature in the giving out of sure fish, crustaceans, and further organisms in rivers. The maximum concentration of magnesium in the groundwater circulatory or unequal heartbeat, low blood pressure, confusion, slowed breathing, unconsciousness, and death is produced in humans. The magnesium range from 3 mg/l to 41mg/l in the study area (Fig.16).

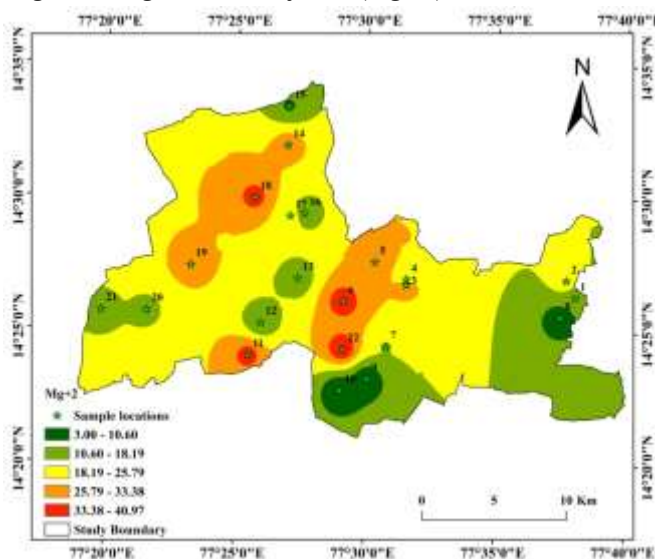


Figure 16: Distribution of Mg^{2+} in Groundwater of the study area.

SODIUM

Sodium is the majorly abundant element on Earth and is broadly dispersed in soils, plants, water. Sodium-ion is everywhere in water because of the high solubility of many sodium salts. A maximum concentration of sodium in groundwater reasons sickness, vomiting, and convulsions muscular twitching, cerebral. The allowable limit is 200mg/l (WHO 2011). The sodium concentration is varied from 98 to 276mg/l. The maximum values distinguished in Thogarakunta and the minimum values distinguished in Vepakunta. The Na^+ concentration is shown in a spatial distribution map Fig 17. In this map shown the Na^+ ranges are within the acceptable limit.

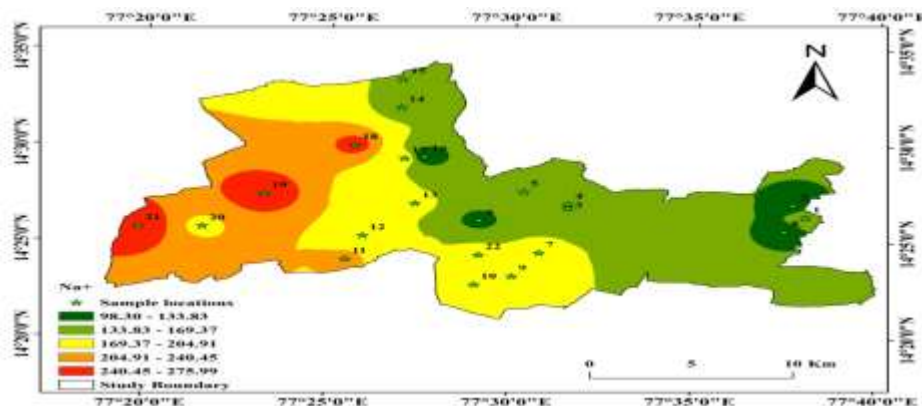


Figure 17: Distribution of the Na⁺ in Groundwater of the study area.

POTASSIUM

Potassium is a vital element for humans. A maximum concentration of potassium indicates a laxative effect. It reasons diabetes and coronary artery diseases, adrenal insufficiency, and heart problems. In the study area distinguished potassium values show a discrepancy from 6 to 7.8mg/l. The permissible limit is 1-8mg/l (WHO 2011). The minimum values distinguished in Ellakuntla village and high potassium values noted in Somavarandlapalli. The K⁺ values are shown in a spatial distribution map Fig 18. From the map shown the K⁺ ranges are within the permissible limit.

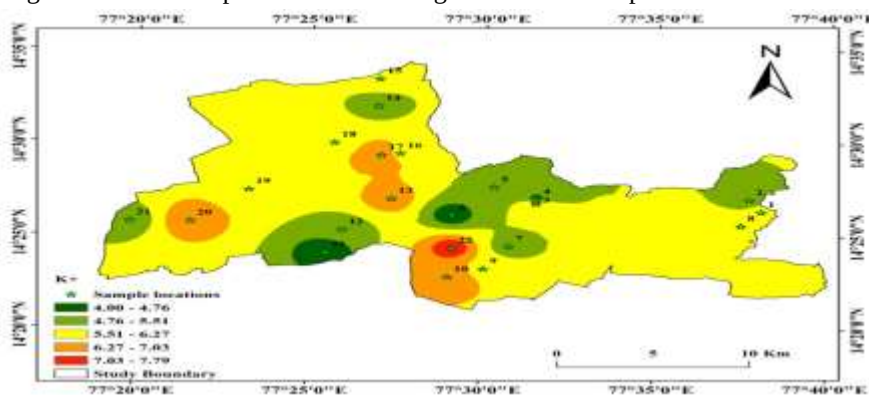


Figure 18: Distribution of K⁺ in Groundwater of the study area

CONCLUSION

In the study area groundwater drawn from 22 bore wells were analyzed for their chemical contents. The analytical results of physical and chemical parameters of groundwater were compared with the standard guideline values recommended by the World Health Organization (WHO, 2011) for drinking purpose. Hydrochemically the ground water contains higher concentrations of TDS and Na⁺, moderate concentrations of Ca⁺ and K⁺ and lower concentrations of Cl⁻, HCO₃⁻, and Mg⁺. Ionic concentration falls in the order Cl⁻ > Na⁺ > HCO₃⁻ > Ca⁺ > Mg⁺ > K⁺ in the groundwater. Assessment of the quality of the groundwater from 22 bore wells indicate that the groundwater belong to hard category and groundwater from majority of the bore wells of the study region is suitable for drinking purposes. Present authors believe that the chemical weathering of the rocks, open sanitation and agricultural return flow have contributed greatly for the major elements of the groundwater. This preliminary study calls for continuous monitoring of the quality of the groundwater in the region as further exploitation of groundwater may increase the values of the some of the parameters viz., EC, TDS, and Mg²⁺ deteriorate the water quality in near future which ultimately will prove to be disastrous for the entire living beings in the region. Spatial distribution maps of pH, Total Hardness, Total dissolved solids; Electrical conductivity, Chloride, Bicarbonate, Sulphate, Calcium, Magnesium, Sodium, and Potassium parameters prepared from the hydrochemical data in GIS environment is useful in assessing the best groundwater quality zone in the study area.

ACKNOWLEDGEMENTS

The first author P. Ravi Kumar, greatly thankful to Department of Science and Technology (DST), Government of India, for financial support through Inspire programme (Sanction order No. DST/INSPIRE Fellowship/2018/IF180877) and also thankful to the Department of Geology, Yogi Vemana University, for necessary facilities for carrying out my research work.

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